Harmonic Chern representatives at the realizable Einstein calibration

23 September 2026. This application starts with a metric and an actual conformally solvable Spin^c structure. No auxiliary form, scalar band, spectral gap, or second-order budget is assumed. Its conclusion is not another Einstein obstruction: the specified harmonic Chern representative is quantitatively close to a bounded almost-parallel form and to low connection-Laplacian modes.

1. Raw hypotheses and statement

Let (M,g) be a closed connected oriented smooth four-manifold, normalized relative to the target $\mathrm{Ric} = -g$. Fix a Spin^c structure with real first Chern class a, whose unperturbed Seiberg–Witten equations are solvable for every metric in $[g]$. The structure may depend on g, but remains fixed when the conformal weight varies. Write

$$V = \mathrm{Vol}(g), \quad p = (a_g^+)^2 > 0, \quad r = \fint |\mathrm{Ric}+g|^2, \quad b = \frac{(-a^2)_+}{V}, \quad w = \sqrt{6}|W^+|,$$

$$\delta = \frac{3}{2}\int (s^2 + 2w^2) - 72\pi^2 p \geqslant 0, \qquad \kappa = r + \delta/V. \tag{1}$$

Ricci uses tensor norms, forms use exterior-algebra norms, and Weyl operators use Hilbert–Schmidt norms. The deficit equals $144\pi^2 D_{\mathrm{LB}}$, or the parameter $\beta = 2$ in `line1_functional_pencil.md`. Assume $\kappa \leqslant 1/2880$.

Let H be the harmonic self-dual representative of a_g^+, with $\|H\|_2^2 = p$, and set $\zeta = -2\pi H$. Our curvature convention is $\mathrm{proj}_{\mathscr{H}^+}(iF_A^+) = 2\pi H$. Put $G(\kappa) = 13\kappa + 4\sqrt{2\kappa}$.

Theorem. For each $0 < \eta \leqslant 1/4$, there is a smooth actual weighted-monopole form $\omega \in \Omega^+(M)$ such that

$$\|\omega\|_\infty \leqslant 2\sqrt{3}, \qquad \boxed{\fint |\omega - \zeta|^2 \leqslant 49\kappa + \eta,} \tag{2}$$

$$\boxed{\fint |\nabla\omega|^2 \leqslant G(\kappa) + \eta, \qquad \fint |d\omega|^2 \leqslant \frac{13}{2}\kappa + \eta.} \tag{3}$$

Define

$$A_{\omega,1/2} = \frac{1}{2}\left(\frac{|\omega|^2}{3} I - \omega \otimes \omega\right),$$

$$d_\omega = \fint \left(|\mathrm{Ric}+g|^2 + |W^+ - A_{\omega,1/2}|^2 + (|\omega|^2 - 2)^2\right), \quad B_\omega = \fint |\Delta_H \omega|^2.$$

The same smooth form satisfies, with universal constants,

$$d_\omega \leqslant 720\kappa + \eta, \qquad B_\omega \leqslant C(1 + \kappa + b), \qquad (1 + B_\omega) d_\omega \leqslant C(1 + \kappa + b)(\kappa + \eta). \tag{4}$$

Here $d\omega$ is exterior differentiation and Δ_H is the nonnegative Hodge Laplacian. The first-order conclusions (2)–(3) do not depend on b; that cost enters only (4).

Let $\mathscr{L} = \nabla^*\nabla$ on Λ^+, let $\mu_0 \geqslant 0$ be its bottom eigenvalue, including zero, and let $\Pi_{\leqslant\Lambda}$ project onto eigenvalues at most $\Lambda > 0$. Then the input harmonic representative satisfies

$$\boxed{\mu_0 \leqslant \frac{G(\kappa)}{2 - \sqrt{720\kappa}}, \qquad \fint |(I - \Pi_{\leqslant\Lambda})\zeta|^2 \leqslant 98\kappa + \frac{2G(\kappa)}{\Lambda}.} \tag{5}$$

No scalar or Hodge spectral gap, scalar-curvature sign, noncollapse, Sobolev constant, or full-curvature bound is used. In particular (2) does not follow from small exterior derivative via a uniform Hodge inverse; its harmonic class comes from the actual curvature equation.

2. Producing the bounded form and its zeroth-order budget

Put $q = -s + w$, $e = s + 4$, $z = s + 2w$, and $q_\pm = \max(\pm q, 0)$. The signed monopole estimate gives $\Delta_+ := \int q_+^2 - 72\pi^2 p \geqslant 0$. Completing the square,

$$\delta = \Delta_+ + \int q_-^2 + \frac{1}{2}\int z^2, \qquad q - 6 = (z - 3e)/2, \quad w - 2 = (z - e)/2. \tag{6}$$

Since $e^2 \leqslant 4|\operatorname{Ric} + g|^2$,

$$⨍ (q-6)^2 \leqslant \delta/V + 18r \leqslant 18\kappa, \qquad ⨍ (w-2)^2 \leqslant \delta/V + 2r \leqslant 2\kappa. \tag{7}$$

Set $Q = \max(q, 3)$ and $f_0 = \max\{\sqrt{Q}, (-s)_+/4\}$. Choose smooth majorants $f_j \geqslant f_0$ converging uniformly to f_0. Conformal solvability supplies

$$D_A\Phi = 0, \quad -iF_A^+ = f_j\sigma(\Phi), \quad \psi = 2\sqrt{2}\sigma(\Phi), \quad \rho = |\Phi|^2 = |\psi|, \quad u = 3\rho/2, \quad \omega = \sqrt{3}\psi/2. \tag{8}$$

With $|\sigma(\Phi)|^2 = |\Phi|^4/8$, the maximum principle gives $\rho \leqslant 4$ and $u \leqslant 6$.

We record the moment inputs, including their truncation cost. Write

$$\begin{aligned} D_0 = {} & 2\int |d\rho|^2 + \int (4\rho|\nabla_A\Phi|^2 - |\nabla\psi|^2) \\ & + 2\int |d\psi|^2 + 2\int E_\lambda, \qquad E_\lambda = \langle W^+\psi, \psi\rangle + w\rho^2/3 \geqslant 0. \end{aligned}$$

All its summands are nonnegative, and $27D_0/8 = \int (qu^2 - f_j u^3)$. For $C = \int Qu^2$, $Z = \int \sqrt{Q}\, u^3$, $E_Q = \int Q^2$, and $C_{\mathrm{act}} = \int f_j^2 u^2$,

$$C - Z \geqslant 27D_0/8, \quad C \leqslant E_Q^{1/3} Z^{2/3}, \quad C_{\mathrm{act}} \geqslant 72\pi^2 p, \quad C_{\mathrm{act}} - C \leqslant \frac{81}{4}\int e^2 + o(1).$$

The last estimate uses $Q^2 - q_+^2 \leqslant 9e^2$ and $f_0^2 - Q \leqslant 9e^2/16$. Consequently, with $S = \Delta_+ + 117\int e^2/4 \leqslant 117\kappa V$,

$$D_0 \leqslant 4S/27 + o(1), \quad \int Q(u - \sqrt{Q})^2 \leqslant S + o(1), \quad 0 \leqslant C_{\mathrm{act}} - 72\pi^2 p \leqslant S + o(1). \tag{9}$$

These are the truncated Hölder moment estimates proved in the functional-pencil note, Section 3, and Round 4, Section 4. All errors here concern weight approximation on a fixed (g, a); no limiting spinor or common smoothing rate for a family is assumed.

The explicit zeroth-order bound follows without an eigenline choice:

$$⨍ (u^2 - 6)^2 \leqslant [234(1 + 2\sqrt{3})^2 + 36]\kappa + o(1) \leqslant 4800\kappa + o(1), \tag{10}$$

because $|u^2 - Q| \leqslant (1 + 2\sqrt{3})\sqrt{Q}|u - \sqrt{Q}|$, and clipping does not increase distance from six. Since $|\omega|^2 = u^2/3$, its amplitude error is at most $4800\kappa/9 + o(1)$. The exact trace-free matrix identity is

$$|W^+ - A_{\omega,1/2}|^2 = \frac{1}{6}(w - u^2/3)^2 + \frac{3}{4}E_\lambda. \tag{11}$$

By (7), (9), and $\int E_\lambda \leqslant 2S/27 + o(1)$, its normalized integral is at most $(2/3 + 4800/27 + 13/2)\kappa + o(1)$. Adding amplitude and Ricci errors gives $d_\omega \leqslant 720\kappa + o(1)$.

3. Differential and harmonic information

The self-dual Weitzenböck formula and (9) give directly

$$\begin{aligned}\int |\nabla\psi|^2 &= 2\int |d\psi|^2 + 2\int E_\lambda - \tfrac{1}{3}\int z\rho^2 \\ &\leqslant 4S/27 + (16/3)\sqrt{V\int z^2} + o(1).\end{aligned} \tag{12}$$

Multiplying by 3/4 proves (3). This controls the full covariant derivative, without a Hessian estimate or Chern-square budget.

The actual curvature has harmonic projection $\operatorname{proj}_{\mathscr{H}^+}(f_j\psi) = -4\sqrt{2}\pi H$. Thus

$$\|f_j\psi + 4\sqrt{2}\pi H\|_2^2 = \tfrac{4}{9}(C_{\mathrm{act}} - 72\pi^2 p) \leqslant 4S/9 + o(1). \tag{13}$$

Moreover $(f_0 - \sqrt{Q})^2 \leqslant f_0^2 - Q$, so

$$⨍ |f_j - \sqrt{6}|^2 \leqslant \tfrac{9}{2}r + \frac{2(\delta/V + 18r)}{(\sqrt{3}+\sqrt{6})^2} + o(1) \leqslant 9\kappa + o(1). \tag{14}$$

Use the identity

$$2\sqrt{2}(\omega + 2\pi H) = (\sqrt{6} - f_j)\psi + (f_j\psi + 4\sqrt{2}\pi H)$$

and $|\psi| \leqslant 4$. The squared triangle inequality gives

$$⨍ |\omega + 2\pi H|^2 \leqslant 4⨍ |f_j - \sqrt{6}|^2 + S/(9V) + o(1) \leqslant 49\kappa + o(1), \tag{15}$$

which identifies the prescribed harmonic Chern representative.

4. Second-order budget and smoothing

Equations (6)–(7) bound the normalized squared integrals of Ric, s, W^+ by $C(1+\kappa)$, and the cutoff bounds $⨍ f_j^2 \leqslant C(1+\kappa) + o(1)$. Hence

$$\int |F_A|^2 = 2\int |F_A^+|^2 - 4\pi^2 a^2 \leqslant CV(1 + \kappa + b) + o(1).$$

The positive-spinor connection curvature is controlled by $W^+, s, \mathrm{Ric}^\circ, F_A$; W^- does not act on this representation. Apply the inherited bounded-section estimate

$$\|\nabla^2 v\|_2^2 \leqslant 2\|\nabla^*\nabla v\|_2^2 + 32\|v\|_\infty^2\|\operatorname{Ric}\|_2^2 + 33\|v\|_\infty^2\|\mathscr{F}\|_2^2, \quad \|\nabla v\|_4^2 \leqslant 4\|v\|_\infty\|\nabla^2 v\|_2$$

to Φ, whose rough Laplacian is controlled by $(s + f_j\rho)\Phi$. Twice differentiating the parallel quadratic map and using the self-dual Hodge formula gives $B_\omega \leqslant C(1+\kappa+b)$. This is the second-order argument in the functional-pencil note, Section 5, and Round 21, Section 2; it differentiates neither the weight nor the Ricci error.

For each fixed metric and class, select one sufficiently large j so that all normalized approximation errors meet their allotted fractions of η. All estimates hold for the same smooth actual solution. The second-order bound holds for every sufficiently close majorant. This proves (2)–(4), with no limiting-solution claim.

5. Connection spectrum and smooth approximate projections

The amplitude bound implies $⨍ |\omega|^2 \geqslant 2-\sqrt{720\kappa+\eta} > 1$, which gives the Rayleigh bound on μ_0. The spectral theorem gives

$$\|(I-\Pi_{\leqslant\Lambda})\omega\|_2^2 \leqslant \|\nabla\omega\|_2^2/\Lambda.$$

By (2)–(3), the normalized tail of ζ is at most $98\kappa+2G(\kappa)/\Lambda+2\eta(1+1/\Lambda)$. The harmonic representative and the spectral projection are fixed as $\eta \downarrow 0$, proving (5) without convergence of spinors.

For a sequence with $\kappa_j \to 0$, even if $b_j \to \infty$, $⨍ |\zeta_j|^2 = 4\pi^2 p_j/V_j \to 2$. Choose $\Lambda_j \to 0$ with $G(\kappa_j)/\Lambda_j \to 0$. Then all its normalized L^2 mass lies asymptotically in the low connection-Laplacian subspace. This does not assert a pointwise bound on H_j, nor is it a scalar-Laplacian cluster statement.

The same form defines $P_\omega = \omega\otimes\omega/2$. Algebra gives

$$\begin{aligned} ⨍ |W^+ + P_\omega - I/3|^2 &\leqslant d_\omega, \\ ⨍ |P_\omega^2 - P_\omega|^2 &\leqslant 9d_\omega, \\ ⨍ |\nabla P_\omega|^2 &\leqslant 12(G(\kappa)+\eta). \end{aligned} \tag{16}$$

Outside $\{|\omega|^2 \leqslant 1\}$, of relative volume at most d_ω, the endomorphism $\omega\otimes\omega/|\omega|^2$ is a rank-one projection with derivative energy at most $2V(G(\kappa)+\eta)$. This does not remove the exceptional set or give an everywhere nonvanishing closed form. An L^2 approximation also does not permit substituting the potentially unbounded ζ for ω in the quadratic Weyl model.

6. An actual family outside every fixed negative scalar band

Take a closed negative Kähler–Einstein surface (M,h), with $\mathrm{Ric}_h = -h$, carrying an actual canonical monopole class; a product of two closed hyperbolic surfaces is an example. In normal coordinates centered at x_0, choose $\phi \in C_c^\infty(B_1(0))$ with $\phi(0)=0$, $d\phi(0)=0$, and $\Delta_{\mathrm{Eucl}}\phi(0) > 0$, for the nonnegative-Laplacian sign convention. Set

$$f_\varepsilon(x) = \varepsilon\phi(x/\varepsilon), \qquad g_\varepsilon = e^{2f_\varepsilon}h,$$

extending the bump by zero outside its coordinate ball. In dimension four,

$$\int |\nabla_h^2 f_\varepsilon|^2 = O(\varepsilon^2), \quad \int |df_\varepsilon|^4 = O(\varepsilon^4), \quad \|f_\varepsilon\|_\infty = O(\varepsilon).$$

The conformal Ricci formula, and cancellation of conformal factors in the four-dimensional squared norm integral of a covariant two-tensor, give $\int |\mathrm{Ric}_{g_\varepsilon}+g_\varepsilon|^2 = O(\varepsilon^2)$. Volumes converge to $\mathrm{Vol}(h)$. Both p and $\int |W^+|^2$ are conformally invariant. Since

$$s_{g_\varepsilon} = e^{-2f_\varepsilon}(-4+6\Delta_h f_\varepsilon - 6|df_\varepsilon|^2),$$

integration of $\Delta_h f_\varepsilon$ gives

$$\int s_{g_\varepsilon}^2 - \int s_h^2 = 36\int(\Delta_h f_\varepsilon - |df_\varepsilon|^2)^2 + 48\int |df_\varepsilon|^2 = O(\varepsilon^2).$$

Thus $\delta_\varepsilon = O(\varepsilon^2)$ and $\kappa_\varepsilon \to 0$, while

$$s_{g_\varepsilon}(x_0) = -4 + 6\varepsilon^{-1}\Delta_{\mathrm{Eucl}}\phi(0) \longrightarrow +\infty.$$

The application therefore genuinely includes metrics outside every fixed negative scalar band. These are integral-error examples, not pointwise Ricci-pinched Yau26 inputs, and not new smooth topologies.

7. Increment and scope

The supplied form lies at the allowed Round-21 parameter $t = 1/2$, where spectral non-saturation gives no contradiction. The moment and Hessian mechanisms are inherited. Their all-scalar composition removes the scalar band from the earlier actual-form supply. No scalar spectral gap is needed when integral Ricci error is an input; this does not remove the spectral-gap term from the different theorem deriving Ricci control from period data alone. Identity (12) also removes the negative-square cost from its first-order and low-mode conclusions. The useful added output is the canonically identified harmonic alignment and spectral/projection information, not a new topological obstruction or existence theorem.

At zero defect this recovers the realizable negative Kähler–Einstein equality case. At positive defect it supplies quantitative integral structure, not global Kähler integrability or geometric compactness.

Proof sources and verification

- `line1_functional_pencil.md`, Sections 2–5: actual all-scalar form production, specialized to parameter two.
- `round4_rigidity.md`, Sections 2–4: cutoff and moment estimates.
- `round21_analytic.md`, Section 2: the bundle Hessian estimate.
- `line1_scalar_band_remainder.md`, equations (9)–(10): earlier scalar-band differential and weighted harmonic remainders.
- `line1_composed_application_check.py`: exact finite coefficients and tensor identities, not certification of gauge solvability or PDE.